Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2402)

Key

SECTION – I

1. D	7. B	13. D	19. C
2. A	8. B	14. A	20. 4213
3. B	9. B	15. B	21. C
4. A	10. C	16. D	22. 2413
5. D	11. C	17. 4213	23. A
6. A	12. D	18. C	24. A

SECTION – II

1. B	6. 17000	11. 10	16. 67
2. C	7. B	12. 216	17. A
3. D	8. 90000	13. A	18. 41
4. D	9. A	14. C	19. A
5. A	10. A	15. 192	20. B

SECTION – III

1. B	7. C	13. D	19. 150
2. C	8. D	14. C	20. B
3. 315	9. 2	15. B	21. 500
4. D	10. D	16. D	22. 309
5. 40000	11. 161	17. B	
6. 10	12. D	18. D	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 539

1. Option A: This is true as realpolitik believed in the politics that served the country's practical interests (raison d'état - a purely political reason for action on the part of a ruler or government) first irrespective of the ethical considerations and that led to several wars, the author says. Hence, Option A is something the author will approve of. Option A is not the answer.
Option B: Many political thinkers who pursued Machiavelli's principles favoured using evil as a legitimate means for political ends. That is how the principle of 'double ethics' was born too (what's ethical for the state as opposed to the individual or in the traditional way). So, it can be said that morality didn't prevent policymakers from believing that using evil is justified as a means to political ends. Option B is not the answer.
Option C: From 'The tension between expediency and morality lost its validity in the sphere of politics', we can understand that expediency (whatever was more convenient and faster to accomplish) was favoured over morality. Hence, there wasn't any tension between the two. This was a feature of the 'realpolitik'. Option C is not the answer.
Option D: This is definitely not true, as understood from the sentence 'These concepts, along with the belief in the superiority of Germanic culture, served as weapons with which German statesmen,' which shows that the belief in superiority of culture along with realpolitik led to German policies. However, the belief in superiority is unrelated to realpolitik. It is an entirely different factor. Hence, Option D is the answer.
Choice (D)

2. Option A: This is apt and the reason why the author had mentioned all these examples. The preceding sentence establishes that Machiavelli is held responsible. Hence, these examples were given to show that they too were opposed.

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Yet, it is Machiavelli's principles that prevailed, unlike the others which opposed the conventional view on politics and ethics. Option A is the answer.
Option B: While the author mentions this directly, the other examples in no way relate to this statement or add or take away weight from it. Hence, this cannot be the purpose of mentioning the other examples. Option B is not the answer.
Option C: While this may be partially true, the purpose of the author mentioning the others was not to show that Machiavelli's principles prevailed because they were the 'need of the hour'. Yes, Machiavelli's thoughts prevailed unlike the others but it is not conclusive to say that those thoughts were the need of the hour and others weren't. Option C is not the answer.
Option D: We are not sure if the aforementioned examples justified evil as a political tool or not. We only know Machiavelli did it. Others have been mentioned to oppose the traditional view on politics and ethics. So, this option is an extreme extrapolation. Option D is not the answer.
Choice (A)

3. Option A: This statement seems to justify doing away with ethics when it comes to the realm of politics, as politics doesn't fit with ethical values in the traditional sense. This is an argument Machiavelli is likely to give to the idealists. Option A is not the answer.
Option B: This option asserts that ethical values determine the ends and limits of politics and come before political values (antecedent authority). In other words, ethics comes above politics. This would be the argument of the idealists. Hence, Option B is the answer.
Option C: This argument shows that morality works in the personal sphere and not necessarily in the political sphere. That is something a realist would argue against an idealist, and is hence, possible for Machiavelli to use in a debate against political idealists. Option C is not the answer.
Option D: Ethics doesn't provide an action plan in the world of politics where there is contestation and coercion. That is a political realist's argument against that of an idealist. This can be said by Machiavelli in a debate against political idealism. Hence, Option D is not the answer.
Choice (B)

4. It is important to understand here that 'a higher type of morality' refers to double ethics/realpolitik/or Machiavelli's realism – state above ethics. So, we are looking for an option that doesn't really portray unethical practice of a state for its own good.
Option A: This represents a scenario where a country is in bilateral agreements with two different countries. The relationship between those two countries has got nothing to do with a country's individual trade agreements. So, it is not unethical on their part. This option doesn't demonstrate a state over ethics kind of scenario. Option A is the answer.
Option B: Attacking another country for resources needed for world dominance is a simple case of state over general ethics. Hence, this is a good example of higher morality. Option B is not the answer.
Option C: This is putting a country's safety before its friendship and its own promises, which is the practical political philosophy that Machiavelli wished to pursue. Personal ethics (in case of friendships/assurances) doesn't necessarily hold when it comes to what's good for the state. Option C is not the answer.
Option D: Freedom of citizens and their rights versus the powers of the state – this debate comes under the purview of the higher morality. In the passage, we understand that a higher morality is putting the state ahead of ethical considerations (the freedom and privacy of citizens should be guaranteed by the government ethically speaking). So, Option D is not the answer.
Choice (A)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 541

5. Option A: We cannot really infer from the passage that Tolkien is the first to invent a fictional language. He 'only moved the goalposts' – it means that Tolkien had extended what had already existed. The author does talk about fictional languages in the first para as well. Option A is not the answer.
Option B: The author does credit Tolkien for going to the extent of developing root words, for his tree of fictional languages, much like the real-world languages. However, we can only infer that he had exceeded all in the depth of his language-development. We cannot say or imply that no other writer had tried doing something similar – creating a tree of languages. Option B is not the answer.
Option C: While Tolkien did transform the idea of fictional languages, it cannot be implied that prior to Tolkien, the creation wasn't as acceptable a practice as it was later. The word 'acceptable' makes this option easy to eliminate, since we aren't really talking about acceptance here. Fictional languages had been used earlier as well, just not as extensively, probably or with the depth that Tolkien introduced. Option C is not the answer.
Option D: Tolkien moved the goalposts considerably. This idiomatic expression shows that what he did changed the dimensions of what existed. In other words, his contribution changed how much fictional languages were considered important for fantasy worlds. Also, he turned invented languages into a 'hallmark' for modern fantasy worlds, showing that what he did made/entrenched the idea/usage of fictional languages more firmly into the modern fantasy worlds. Option D is the answer.
Choice (D)

6. Option A: The Sapir-Whorf hypothesis argues that our world view, perspective, thoughts, and way of thinking change based on the language we use. In other words, language changes how we go about our thought processes. So, if we can pick the alien language and learn to change how to think from that, it proves the hypothesis true. Hence, this option strengthens the hypothesis that language determines/influences thought. Option A is the answer.
Option B: While this option shows a correlation between language and behaviour/thinking (pejorative – slang, disrespectful), it doesn't establish a causal relationship. We cannot be sure whether language leads to niceness or niceness leads to a language with few(none) pejorative terms. Option B is not the answer.
Option C: This option shows that there is a connection between language we speak and thought process. Yet, it cannot strengthen the Sapir-Whorf hypothesis because it doesn't show that language influences thought. Hence, Option C is not the answer.
Option D: Whether the words in such languages make the people feel those emotions, or whether the words exist because people feel those emotions and want to have words for them is not clear from the option. Hence, Option D is not the answer.
Choice (A)

7. Option A: While Nadsat does create an estrangement effect (a sense of alienation or a certain disconnect), the example was not picked to describe the effect. It was to show how languages influence the way we think. In this case, Nadsat makes readers think and observe things in a certain way. Option A is not the answer.
Option B: This is the summation of Sapir-Whorf hypothesis, which the author illustrates further using the example of Nadsat. This can be inferred from the starting phrase of the para 'In a similar vein'. So, the purpose of the Nadsat example is to show how language can influence our thought (in this case, it creates an estrangement effect, for example). Option B is the answer.
Option C: The author doesn't really talk about the relevance of Nadsat or about the longevity of invented languages (for that matter, longevity of invented languages has not been discussed anywhere in the passage, and not just in this example). Yes, the Nadsat has 'aged well' the author says, but the question is more to do with why Nadsat was mentioned in the first place. Option C is not the answer.
Option D: Neither is the focus on specific invented words, nor did the author specifically want to discuss violence. Yes, Nadsat created an astonishing effect around the chilling violence depicted in the book. However, the purpose of mentioning Nadsat is far removed from the violence in the story. And not all invented languages 'distort' the reader's perception (and that too towards violence specifically). Hence, Option D is not the answer.
Choice (B)

8. Option A: This is true, as can be understood from Tolkien's invented languages which followed the 'tree of languages' model – branching off to form individual languages. Tolkien's languages followed the model of real languages as understood from the sentence '*Following the Indo-European model… Tolkien started by…over time*'. Option A is not the answer.
Option B: The fictional languages weren't developed to demonstrate the ability of language to manipulate us. Yes, language does, and yes, there is a school of thought that studies how languages manipulate us. However, that isn't the purpose of the art-langs. Hence, Option B is the answer.
Option C: The first line of the passage mentions that art-langs '*give depth and verisimilitude to imaginary worlds in works of fiction*'. From this, we can understand that invented languages lend authenticity (verisimilitude) to the words created by the authors, that they offer depth to the fictional worlds, and increase the sense of believability. Option C is not the answer.
Option D: Some of the invented languages like Vril-ya engage with contemporary linguistics, the passage mentions. Also, Tolkien's invented languages follow the same model of evolution as languages evolving from a proto-language do. So, yes, we can infer that invented languages follow similar rules. This is further strengthened by the passage mentioning that Vril-ya '*supposedly underwent various phases of development, corresponding to the theories of the nineteenth-century Oxford philologist Max Müller.*' Hence, Option D is not the answer.
Choice (B)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 489

9. In the first para of the passage, the author mentions that "health economists generally assume that the demand for health care derives from a demand for health". To weaken this assumption, an option that talks about how demand for health is not the only driving factor, will be the correct answer.
Option A: This option talks about consumers choosing insurance that covers chronic, acute illnesses rather than smaller ailments. This does not weaken the assumption that health economists make since the consumer's primary motivation is demand for health. Option A is incorrect.
Option B: In this, the consumer considers health insurance not only for health purposes but also for financial benefits such as tax exemptions. This weakens the health economists' assumption that health care demand is derives from a demand for health. Hence, Option B is the correct answer.

Option C: This option talks about the income of a consumer being a factor in determining preferences and the extent of investment when it comes to health care. But that does not weaken the assumption that the demand for health care derives from a demand for health. Hence, Option C is incorrect.
Option D: This option also connects health choices with earning capacity and price, which is consistent with the assumption spoken about in the para. Option D is not the answer.
Choice (B)

10. Option A: From the first line of the 4th para – "Economic theory also posits that health care demand is jointly determined with supply" – we see that the former part of this option is incorrect. Option A is not the answer.
Option B: In the 3rd para of the passage, the author uses the phrases "it is believed that they are correlated with otherwise unmeasured preferences" and "interpretation is difficult" with respect to sociodemographic characteristics in empirical models. Therefore, it cannot be said that empirical models shed light on direct and proxy determinants. Option B is incorrect.
Option C: From the second part of the 3rd para, we see that empirical models often include sociodemographic characteristics such as "age, ethnicity, sex, and education" to understand consumer preferences when it comes to healthcare use. Theoretical models rarely do the same. Option C is the correct answer.
Option D: While the former part of this option can be understood from the second para of the passage, it would be a stretch to state that empirical models address this drawback. We can only understand, from the passage, that empirical models include sociodemographic characteristics to determine demand. From this, we cannot glean if the drawback is addressed. Option D is not the answer.
Choice (C)

11. The passage states that economic theory does not take into account how individuals form preferences, while determining health care demand. This is because people "vary in attitudes and preferences toward risk, willingness to trade-off better health tomorrow for increased consumption today, and likes and dislikes".
Option A: The passage does not mention the relation between health care price and the provider of the service. It does not discuss this aspect and economic theory may or may not make this assumption to account for determining health care demand. Hence, we cannot comment whether this assumption is likely or not likely.
Option B: The passage mentions that "Economic theory also posits that health care demand is jointly determined with supply". From this, we can infer that, according to economic theory, the demand for health care is related to the supply.
Option C: The passage mentions that "economic theory is agnostic as to how individuals form preferences". This option presents one of the ways in which individuals form preferences. Hence, economic theory is not likely to make this assumption while determining health care demand. Individuals may take a health care plan to protect themselves from chronic illness, but economic theory doesn't concern itself with how individual specific preferences are made. Of the four options, economic theory is least likely to make the assumption related to how individuals form preferences, i.e., option C. Hence, Option C is the correct answer.
Option D: The passage mentions that "Health care prices and income too are almost never observed in the way we would like". This implies that this observation is important for economic theory while determining demand. We cannot infer exactly *how* the health care prices and income play a part but we can definitely infer *that* the two aspects play a part.
Choice (C)

12. Option A: The author mentions in the second para of the passage that economic theory does not take into account how individuals form preferences, since there is an element of unpredictability involved which is not easily calculable. However, author mentions that understanding this is vital to determine demand and would definitely recommend this option. This makes Option A an incorrect answer.
Option B: In the last line of the 2nd para, the author states that "Health care prices and income too are almost never observed in the way we would like", which means that it is likely the author would recommend this step. Option B is not the answer.
Option C: In the last line of the passage, the author suggests that "competing concerns about usefulness of particular variables as predictors of health care use and the potential biases they introduce must be confronted". So, this recommendation is possible. Hence, Option C is incorrect.
Option D: At the end of the first para - "Health then, along with price (or its proxy health insurance), income, and consumer preferences, play the main role in formal economic models of consumer decision making about health care." This shows that this has already been incorporated in economic theory and "Yet, this theory takes us only so far". There is no point in exploring an avenue that leads to a dead end. This is not something that the author would recommend, making it the correct answer.
Choice (D)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 557

13. Option A: The windshield phenomenon – people feeling that the number of insects is going down – is a different phenomenon compared to the shifting baseline syndrome, where humans miss out on the gradual changes in the environment that they live in. Hence, Option A is not the answer.

Option B: Pride is probably a bit of an exaggeration. The fishermen may not have been proud, per se, about their catch as compared to their predecessors, since they didn't even compare. It is the author who is comparing the size of the fish, pointing to the reduction in size. Hence, Option B is not the answer.
Option C: The author didn't establish causation. The author has merely pointed out that humans do not remember. The author didn't establish that it is the absence of remembering that is causing the environment crisis of sorts. Hence, Option C is not the answer.
Option D: This is apt as the author uses the smile of the fishermen as a metaphor – to indicate that they didn't even notice the difference in the size of fish. While the fish have grown much smaller, the fishermen continue to be happy, as they don't have a comparison standard with the past. Hence, Option D is the answer.
Choice (D)

14. Shifting baseline syndrome depicts a sense of amnesia, people forgetting what things were like in the past. So, this syndrome has two tenets: that people are happy about something, and secondly, whatever they are happy about is actually on the decline compared to the past. We have to look for these two tenets.
Option A: The local fishermen are celebrating something believing it is a good thing, but there is a decline in the long run, which they are unmindful or ignorant about. This is an example of shifting baseline. Hence, Option A is the answer.
Option B: The younger residents pick a bird that is common in the area. From this we don't know if there is amnesia involved because the same bird may have been common a long time ago as well. The option doesn't clarify that, neither does it show a sense of positivity felt by people despite a negative trend. This is not an example of shifting baseline syndrome. Option B is not the answer.
Option C: Country X has been appreciated for increasing its forest cover. This is in no way a justification of a lower forest cover, nor does it indicate a sense of complacency. It is just appreciation of good work, without comparison. Hence, this is not example of shifting baseline syndrome. Option C is not the answer.
Option D: The country has taken the standard as a population that isn't endangered. They are totally ignoring the possibility that the population could still be remarkably lower compared to the past. At the same time, the option doesn't tell us whether they are actually ignorant or resuming whaling because they want to irrespective of whether the number is low or not. So, neither ignorance nor a sense of celebration have been shown here. This is not an example of shifting baseline syndrome. Option D is not the answer.
Choice (A)

15. Option A: The footprint of tigers has drastically reduced, yes. The author though feels 'reassurance in the survival' ignores the 'value of abundance.' In other words, mere survival shouldn't reassure us, since the focus point here is that tigers are not abundant anymore. However, the author doesn't really exhort the tiger activists here, or tell them what they should do or what they shouldn't. Hence, Option A is not the answer.
Option B: This is what the author is trying to say, that 'abundance' is an important concept, and not just survival. And that is particularly true for large predators, such as the tigers. The value of their abundance shouldn't be ignored. In other words, these predators should be abundant for ecosystems to function. So, the author is trying to indicate that large predators need to be abundant, and just survival isn't the end of the battle. Option B is the answer.
Option C: While low numbers can make a species functionally extinct, we don't know if tigers are functionally extinct as of now. The author doesn't clarify it in the para either. Hence, Option C is a far-fetched hypothesis, and not the answer.
Option D: Loss of habitat hasn't been mentioned in the case of tigers. So, losing focusing on the loss of habitat is an exaggerated hypothesis, and can be easily eliminated. Option D is not the answer.
Choice (B)

16. Option A: This can be inferred from the sentence: 'Large animals, especially top predators like tigers, connect ecosystems to one another'. They do by walking and eating and defecating and dying, the author says. Unless, they are part of multiple ecosystems that cannot be possible. Hence, Option A is not the answer.
Option B: Whale carcasses become entire ecosystems of their own in nutrient-poor conditions. In other words, nutrients are needed for ecosystems to thrive (the whale carcass providing the required nutrients here). If nutrients weren't important for ecosystems, the author wouldn't need to emphasize on this statement: '*sunken whale carcasses form the basis of entire ecosystems in nutrient-poor places.*' Hence, Option B is not the answer.
Option C: From the example of sea otters we can understand that if the prey populations go boom or crash, it affects the entire ecosystem because of a chain reaction. So, this statement can be inferred. Hence, Option C is not the answer.
Option D: While we know that a sunken whale carcass can be the 'basis' of an ecosystem, we don't know and can't say whether a whale (a live one as we understand it) can be an ecosystem of its own. That is too much of an extrapolation, considering we don't understand and the passage doesn't explain how this ecosystem can function per se. Hence, Option D cannot be inferred and is the answer.
Choice (D)

Solution for question 17:

17. Sentence 1 is a grammatically independent sentence that establishes the meaning of equity, in positive terms. It is contextually not independent because it states that equity in literal sense 'expands to'. So, something must have previously been mentioned about equity.

Sentence 2 is also an independent sentence about the scope of equity.
Sentence 3 gives us examples of countries from the UK to India which have incorporated the principles of equity in their legal systems. This sentence is more about specific examples.
Sentence 4 gives us the definition of equity as provided by Aristotle.
We can see that 42 is a block, because 4 gives Aristotle's definition of equity as 'correction of law' and how it is done – 2 explains that equity fills the lacunae/gaps that exist in law. 1 as explained above is downstream to 4. But, it won't make sense to come after the examples in 3. So, 4213 is the order.
Ans: (4213)

Solution for question 18:

18. The statement immediately preceding blank (1) talks about the relation between economy and witches being negative. The statements following blank (1) elaborate on this – attacking livestock, crop failure etc. The highlighted sentence, which talks about some pact, would break the thought flow of the statements if inserted in blank (1). Hence, the highlighted sentence does not fit into blank (1).
The statement immediately preceding blank (2) states that witches did not profit from their magic and the statement immediately following blank (2) explains why they did not. Therefore, these two statements are connected and inserting the highlighted sentence between them would break the flow. Hence, the highlighted sentence does not fit into blank (2).
If the highlighted sentence is not in blank (3) "it simply stated" in the statement immediately following blank (3) would refer to "weather magic" because the statement immediately preceding blank (3) talks about weather magic. But weather magic cannot state anything. If the highlighted sentence and the statement immediately following blank (3) together would mean: The pact with the devil is from a contract like that of Goethe's Faust in the sense that it was about the wishes of the magicians and the pact with the devil was about the witch submitting to the will of the demon, implying it is about the wishes of the demon. These two taken together compare the pact with the devil and a contract like Goethe's Faust. Hence, the highlighted sentence fits into blank (3). "not about the personal interests or the wishes of the witches" in the statement immediately following blank (4) indicates that it is the wish of the demon. Therefore, the statements immediately preceding and following blank (4) talk about destruction and hatred brought about by the demon. The highlighted sentence, which talks about a pact with the devil would break the thought flow if inserted in blank (4). Hence, the highlighted sentence does not fit into blank (4).
Choice (C)

Solution for question 19:

19. Option A: The para doesn't say that there are gaps in stories which are covered by narrative. The para says that stories fill the gaps in our experience. Hence, it is not apt. Option A is not the answer.
Option B: The option is close but there are two issues. Firstly, historians don't use narrative as a consequence of the constraint of chronology and details (like a deliberate solution to the constraint). Secondly, the para doesn't say historical explanations are inept on their own. Such a distinction has not been made between historical explanations. Option B is not the answer.
Option C: This mentions all the ideas of the para. Firstly, narrative is not a perfect replacement for experience. Narrative does fill some gaps, and offers continuity to the story, that is being told by a historian. Thirdly, historians are limited by chronology and accuracy. Hence, this is the correct answer.
Option D: It is not true that historical stories lack in experience. It is definitely not true that whatever we experience through narrative is lacking in anything particularly ***because of constraints of chronology and details***. These are unrelated points of the para. Hence, Option D is easy to eliminate.
Choice (C)

Solution for question 20:

20. Sentence 1 starts with a contrasting connector 'Indeed', followed by something positive about Anarchism.
Sentence 2 refers to some analogy in a previous sentence and equates that too organised activity of free human beings, thus defining Anarchism.
Sentence 3 is an independent sentence just stating an opinion about the current conditions.
Sentence 4 gives us the example of organic growth producing colour and form in flowers.
It is easy to see that 42 is a block, since 4 offers the analogy, the example, a flower compared to how social harmony happens. We can also see that 1 cannot come before 42, because 2 introduces Anarchism. And 3 cannot come between 42 and 1 because the 'indeed' in 1 won't make sense. So, the only possibility is 4213.
Ans: (4213)

Solution for question 21:

21. Option A: This option seems to indicate that cinema cannot equal theatre without the actor's presence, leaving room for the possibility that there could be the presence of an actor in cinema. However, the para rules that out. Cinema cannot pretend to be parallel or equal because it can never offer the presence of an actor, according to the para. Option A is not the answer.
Option B: While the first part is approximate, the second part is not true. The stage welcomes every illusion except that of presence – this cannot be equated to 'every part of the stage is an illusion except presence of an actor.' This is because

there is a difference between saying theatre is an illusion, versus saying, theatre welcomes (accepts, accommodates) an illusion. Option B is not the answer.
Option C: This option offers the main idea of the para: that critics of cinema find theatre pleasurable (because of the actor's physical presence). The sub-idea that cinema is more representative of reality is also present in this option. Option C is a good summary of the para. Option C is the answer.
Option D: While the greater part of the option is not inaccurate, it is not correct to say theatre is elusive in nature. That would mean theatre is not for real, which doesn't reflect the information in the para. (Also, on a side note, please understand that theatre welcomes illusion, indicates what happens in theatre favors illusion. Doesn't mean theatre itself is elusive.) Option D is not the answer.
Choice (C)

Solution for question 22:

22. Sentence 1 is an independent sentence about The Plague.
Sentence 2 is about the opinion of Tony Judt about The Plague. (We can understand that 2 has to come somewhere after 1, not necessarily immediately after.)
Sentence 3 is about the genome 'also' being a testing ground (so we know something similar has been mentioned previously).
Sentence 4 is about what happens in The Plague.
We can understand that 2, 4 and 1 are about The Plague. 2 introduces it (Camus's novel). So, it comes first. 1 and 4 are interchangeable, but then 1 indicates that this is an allegory, and 3 talks about the genome being a testing ground too. So, 1 and 3 cannot be separated. So, 2413 is the order.
Ans: (2413)

Solution for question 23:

23. Option A: This option summarises all the important points of the para. Visualisations aren't getting enough attention from scientists. This is a cause for concern as there is the risk (potential - future) of misunderstanding of science by the public, a misunderstanding caused by wrong information/illustrations. Option A is the answer.
Option B: The para is negative in tone and doesn't really have a positive advice that visualisations must be used by scientists to give the public a better understanding. This is because visualisations are already being used. They are just not being used well and are being incorrect, thus providing misleading information. Option B is not the answer.
Option C: Graphs and diagrams are part of the visuals the author is talking about. So, it is incorrect to say that graphs and diagrams are getting more attention from the authors compared to visualisations. Option C is easy to eliminate. Option C is not the answer.
Option D: The para says 'few scientists' (meaning none) pay attention to visualisations. This option changes the essence completely because the para is critical about the way scientists use visualisations (or the lack thereof of accuracy). Hence, Option D is not the answer.
Choice (A)

Solution for question 24:

24. The highlighted sentence briefly expands on "explore how to exist in the world" by stating that philosophy is an activity: a relationship with the world. "how to exist in the world" in the statement immediately preceding blank (1) therefore links with "relationship towards the world" in the highlighted sentence. Therefore, these two statements are connected. Hence, the highlighted sentence fits into blank (1).
The highlighted sentence cannot fit into blank (2) because the statement immediately following blank mentions "they". From the given context, it can be inferred that "they" refers to "the three volumes of *Philosophie*" and not the highlighted sentence as the highlighted sentence does not have a plural noun. Hence, the highlighted sentence does not fit into blank (2).
The statements immediately preceding and following blank (3) talk about what the three volumes of *Philosophie* entail. The highlighted sentence, which talks about what philosophy is, would break the flow if inserted in blank (3). Hence, the highlighted sentence does not fit into blank (3).
Also, "these stages of existence" in the statement immediately following blank (4) has to refer to "orientation, existence, and metaphysical transcendence" mentioned in the statement immediately preceding blank (4). The highlighted sentence, which talks about what philosophy is, would break the flow if inserted in blank. Hence, the highlighted sentence does not fit into blank (4).
Choice (A)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	5, 8, 16, 18
Difficult	1, 2, 3, 6, 7, 9, 11, 12, 13, 14, 15, 17, 19, 20, 22, 23, 24
Very Difficult	4, 10, 21

SECTION – II

Solutions for questions 1 to 5:

The following table provides the dungeons that will open when each lever is operated once:

	D1	D2	D3	D4	D5	D6	D7
L1				✓	✓	✓	
L2		✓		✓		✓	✓
L3			✓	✓	✓	✓	
L4	✓	✓		✓	✓	✓	

From (ii), if L1 and L2 are operated, all the dungeons will be open. But it is given that only four persons can escape. Hence, this is not possible.
If L1 and L3 are operated, two dungeons, D1 and D4, will be opened. Hence, this is also not possible.
If L1 and L4 are operated, four dungeons, D2, D3, D4 and D6, will be opened. Hence, this case is possible. From (ii), Adam, Bob, Evan and Frank must be in these four dungeons.
From (i), if L2 and L1 are opened, all the dungeons will be opened. Hence, this is not possible as there must be at least two persons who cannot escape.
If L2 and L3 are opened, five dungeons, D2, D3, D5, D6 and D7, will be opened. Since D1 and D4 are not opened, this is possible.
If L2 and L4 are opened, three dungeons, D1, D5 and D7, will be opened. By observation, we can see that these three dungeons are all different from the four dungeons that Adam, Bob, Evan and Frank are in. Hence, these three dungeons must have been occupied by Charles, Donald and Gary. However, it is given that Gary cannot escape. Hence, this case is not possible.
Therefore, in (ii), L2 and L3 must have been opened. D1 and D4 must have Evan and Gary. Since Adam, Bob, Evan and Frank are in D2, D3, D4 and D6, and Evan is common to both the lists, Evan must be in D4, while Gary must be in D1. Adam, Bob and Frank must be in D2, D3 and D6, in any order. Charles and Donald must be in D5 and D7, in any order.
From (iii), if L1, L2 and L3 are operated, three dungeons, D1, D2 and D7, will be opened. Hence, this case is possible. However, in this case, both Bob and Frank cannot escape (as Bob and Frank are in two dungeons among D2, D3 and D6 and among D1, D2 and D7, only one dungeon is in common). Hence, this case is not possible.
If L1, L3 and L4 are opened, two dungeons, D2 and D5, will be opened. Hence, this case is also not possible.
If L2, L3 and L4 are opened, five dungeons, D1, D3, D4, D6 and D7, will be opened. Since Bob and Frank are in D2, D3 and D6, and since it is given that Bob and Frank escaped, Bob and Frank must be in D3 and D6 in any order. Adam must be in D2. Also, Donald must be in D7, while Charles must be in D5.
Hence, this is the only case possible and the dungeon in which each person is in is provided in the table below:

Dungeon	1	2	3	4	5	6	7
Person	Gary	Adam	Bob/Frank	Evan	Charles	Frank/Bob	Donald

1. Evan is in D4. Choice (B)

2. Donald is in D7. Choice (C)

3. If each lever is operated exactly once, then D1, D2, D3 and D6 will be closed. In this case, Gary, Adam, Bob and Frank (since both D3 and D6 will be closed) definitely cannot escape. Hence, more than one of the persons given in the options cannot escape.
Choice (D)

4. Operating L3 will open D3, D4, D5 and D6. In this case, both Bob and Frank will be able to escape irrespective of which dungeon they are in. Hence, Frank can be in D3 or D6.
Choice (D)

5. We have already checked for every possible triplet of dungeons except L1, L2 and L4. If L1, L2 and L4 are operated, three dungeons, D3, D6 and D7, will be opened. From the above solution, we have seen that if L1, L3 and L4 are operated, only two dungeons, D2 and D5, will be opened. Charles, who is in D5 can escape.
Choice (A)

Solutions for questions 6 to 10:

In the given table, we can fill up the rows where two values are given (since Cost per tire × Number of tires = Total Cost).
The following table presents the updated information, with each row labelled from A to E for reference:

Row	Cost per toy (in ₹/unit)	Number of toys	Total Cost (in ₹)
A	39		
B	40	16500	6,60,000
C	21	19000	3,99,000
D			7,65,000
E	25	15000	3,75,000

From (ii), number of toys manufactured through Process III is 2000 more than that by Process IV. These two processes cannot be any combination of B, C and E since the difference between no two of them is 2000.
From (ii), Process IV can only be C, D or E (since only these can have the second lowest cost per toy).
From (v), Process III can only be A, B or D. However, from (iii), Process III cannot be D. Hence, Process III can only be A or B.
If Process III is A, Process IV can be C, D or E. If Process III is A, then the number of toys manufactured cannot be greater than $\frac{750000}{39} \approx 19230$ for the total cost to be less than 750000.
Therefore, Process IV cannot be C.
Let Process IV be D. Let the number of toys manufactured through Process IV be x. The number of toys manufactured through Process V will be x + 2000.
Since the average cost for all the five processes combined is 31.8,
31.8 × (x + 2000 + 16500 + 19000 + x + 15000) = 39 × (x + 2000) + 660000 + 399000 + 765000 + 375000
⇒ x ≈ 24695
We can say from this that Process IV cannot be D as well.
Let Process IV be E. Then, the number of toys manufactured through Process III will be 19000. This case satisfies all the conditions.
If Process III is B, Process IV cannot be C and E. Hence, Process IV can only be D. The number of units manufactured through Process IV will be 14500 and the cost per unit will be ₹57.76. However, this will violate condition (ii), as the cost per toy for this process will not be the second lowest.
Hence, Process III is A and Process IV is E.
The total cost for all the five processes combined = 39 × 17000 + 660000 + 399000 + 765000 + 375000 = 2862000
Let x be the number of units manufactured through D.
From the average cost, the total cost = 31.8 × (x + 67500)
The number of units manufactured through D = 22500 and the cost per toy = 34
From (iii), D represents Process V. From (iv), C represents Process II and B represents Process I.
The following table provides the values for each process:

Process	Cost per toy (in ₹/unit)	Number of toys	Total Cost (in ₹)
III	39	17000	6,63,000
I	40	16500	6,60,000
II	21	19000	3,99,000
V	34	22500	7,65,000
IV	25	15000	3,75,000

6. 17000 toys were manufactured through Process III.
Ans: (17000)

7. The total manufacturing cost was the highest for Process V.
Choice (B)

8. The total number of toys manufactured during the given month = 90000.
Ans: (90000)

9. Hari will use Process II for manufacturing 90000 toys in the next month.
Total cost of manufacturing = 90000 × 21 = 18,90,000
Choice (A)

10. The cost per toy for Process II is ₹21.
Choice (A)

Solutions for questions 11 to 15:

From (ii) and (v), we can understand that 72 Senzus = 52 Urmis.
From (i), he had a total of (2/3)*72 = 48 Lingots and (3/2)*52 = 78 Tristans.
Let the number of Senzu coins, Lingot coins, Urmi coins and Tristan coins he has be x, y, y, and (x + 10) respectively (from(iv)).
From (iv), x + y + y + x + 10 = 66 ⇒ x + y = 28.
From (iii) and (vi), we know that he had 7 Tristan Notes and each note is worth 8 Tristans.
The value of the notes he has = 7*8 = 56 Tristans. Therefore, the number of Tristan coins that he had = 78 – 56 = 22 Tristans.
Now, x + 10 = 22 ⇒ x = 12.
We know that x + y = 28. So, y = 16.
Therefore, the number of Senzu coins, Lingot coins, Urmi coins and Tristan coins that he had with him are 12, 16, 16, and 22 respectively.
From (iii), he had 8 Lingot notes.
He had a total of 48 Lingots and 16 Lingot coins. Let the value of each Lingot note be P times the value of each Lingot coin.
16 + 8P = 48 ⇒ P = 4. Therefore, the value of each Lingot note is 4 times the value of each Lingot coin, i.e. 1 Lingot note = 4 Lingots.
From (vi), he had a total of 28 notes, out of which 8 were Lingot notes and 7 were Tristan notes. Let's say that the number of Senzu notes is s and the number of Urmi notes is u.
s + u = 13.
Since he had at least three notes of each currency and from (iii), u cannot be greater than 5. Therefore, the possible values for u are 3, 4, and 5.

If u = 3, s = 10.
In that case the value of each Urmi note would be $\frac{52-16}{3}$ = 12 Urmis and the value of each Senzu note would be $\frac{72-12}{10}$ = 6 Senzus
If u = 4, s = 9.
In that case the value of each Urmi note would be $\frac{52-16}{4}$ = 9 Urmis and the value of each Senzu note would be $\frac{72-12}{9}$ = 6.67 Senzus. But it is given that in each currency, the value of each note is an even multiple of the value of each coin in that currency. Therefore, u cannot be 4.
If u = 5, s = 8.
In that case the value of each Urmi note would be $\frac{52-16}{5}$ = 7.2 Urmis and the value of each Senzu note would be $\frac{72-12}{8}$ = 7.5 Senzus. But it is given that in each currency, the value of each note is an even multiple of the value of each coin in that currency. Therefore, u cannot be 5.
Therefore, the only possible values of u and s are 3 and 10 respectively.
The following is the overall distribution of coins and notes that he had:

	Number of Coins	Number of Notes	Value of a Note/Value of a Coin	Total Value
Senzu	12	10	6	72 Senzus
Lingot	16	8	4	48 Lingots
Urmi	16	3	12	52 Urmis
Tristan	22	7	8	78 Tristans

11. Mohan has 10 Senzu notes.
Ans: (10)

12. An Urmi note is worth 12 Urmi coins. 52 Urmis = 72 Senzus ⇒ 1 Urmi = 72/52 Senzus = 18/13 Senzus
13 Urmi notes = 13 × 12 Urmis = 13 × 12 × 18/13 Senzus = 216 Senzus.
Ans: (216)

13. Mohan had 16 Lingot coins. This is equal to 24 Senzu coins, which, in turn, is equal to 4 Senzu Notes.
Choice (A)

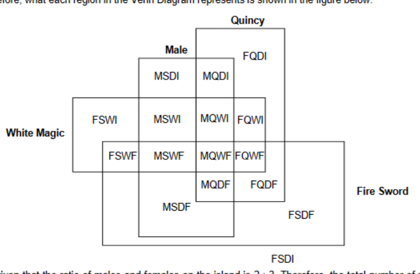
14. 78 Tristans = 48 Lingots => 13 Tristans = 13 * 48/78 = 8 Lingots. 1 Lingot note = 4 Lingots. Therefore, 13 Tristan coins can be exchanged for two Lingot notes. Hence, the total number of Lingot notes and Urmi notes with him 10 + 3 = 13.
Choice (C)

15. He had a total of 48 Lingots with him. Since the total value of notes and coins that he had with him in each currency is the same, the total amount that he had with him will be 48*4 = 192 Lingots.
Ans: (192)

Solutions for questions 16 to 20:

Here, we have four categories – Gender, Tribe, Practice, and Mystical Weapon.
In each category, there are two types, and any islander belongs to exactly one of these types.
In the figure given below, any islander inside the region represented by 'Male' is a male and any islander outside of it is female.
Any islander inside the region represented by 'Quincy' belongs to the Quincy tribe and any islander outside of it belongs to the Shinigami tribe.
Any islander inside the region represented by 'White Magic' practises White Magic and any islander outside of it practises Dark Arts.
Any islander inside the region represented by 'Fire Sword' owns a Fire Sword and any islander outside of it owns an Ice Shield.
Let us represent each region in the following sequence – Gender(M/F)-Tribe(Q/S)-Practice(W/D)-Weapon(I/F). For example, in the figure shown below, MSWF represents Males belonging to Shinigami tribe practicing White Magic and owning a Fire Sword.

Therefore, what each region in the Venn Diagram represents is shown in the figure below:



It is given that the ratio of males and females on the island is 2 : 3. Therefore, the total number of males on the island will be $\frac{2}{5} \times 360$
= 144.
Hence, the value of x in the given figure will be 144 – 123 = 21.
Also, there are a total number of 360 inhabitants on the island. From this, we can calculate the value of y as y = 360 – 317 = 43.

16. The number of islanders who practice Dark Arts and own a Fire Sword can be found out by adding the numbers in the region outside of 'White Magic' and inside 'Fire Sword', i.e. MQDF + MSDF + FQDF + FSDF = 21 + 11 + 13 + 22 = 67.
Ans: (67)

17. The number of islanders who belong to the Shinigami tribe and own an Ice Shield will be MSWI + FSWI + MSDI + FSDI. Since we are not including those who belong to the Shinigami tribe, practice White Magic and own an Ice Shield, we are only adding MSDI + FSDI = 16 + 43 = 59.
The number of islanders who belong to the Shinigami tribe and practice White Magic will be MSWI + MSWF + FSWI + FSWF. Since we are not including those who belong to the Shinigami tribe, practice White Magic and own an Ice Shield, we are only adding MSWF + FSWF = 17 + 27 = 44.
Therefore, the number of islanders belonging to the Shinigami tribe who either own an Ice Shield or practice White Magic but not both is 59 + 44 = 103.
Choice (A)

18. The number of females belonging to the Shinigami tribe and owning Ice Shields = FSWI + FSDI = 47 + 43 = 90.
The number of males belonging to the Quincy tribe and practising White Magic = MQWI + MQWF = 20 + 29 = 49.
Difference = 90 – 49 = 41.
Ans: (41)

19. The total number of females who neither own an Ice Shield nor belong to the Quincy tribe = FSDF + FSWF = 22 + 27 = 49.
Choice (A)

20. The number of males owning Fire Swords can be found out by adding the numbers in the region common to 'Male' and 'Fire Swords' i.e. MSWF + MQWF + MSDF + MQDF = 17 + 29 + 11 + 21 = 78. The number of females owning Ice Shields can be found out by adding the numbers in the region outside of both 'Fire Swords' and 'Male', i.e. FQDI + FSDI + FQWI + FSWI = 33 + 43 + 21 + 47 = 144. Difference = 144 – 78 = 66.
Choice (B)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	16, 19
Medium	1, 2, 17, 18, 20
Difficult	3, 4, 5, 11, 12, 13, 14, 15
Very Difficult	6, 7, 8, 9, 10

SECTION – III

Solutions for questions 1 to 22:

1. Let the average number of runs scored after four innings be k.
Therefore, the total number of runs scored in the first four innings = 4k

It is given that, $\frac{4k+35+46+39}{7} < k$
[∵Average after seven innings was less than that after four innings]
⇒ 3k > 120
Let the number of runs scored in the eighth innings be x.
Again, $\frac{4k+120+x}{8} = 39$
⇒ 4k + x = 192.
Now, since 3k > 120, 4k > 160.
In the above equation, x will be maximum, when 4k is minimum. The minimum possible value of 4k is 161, which gives the maximum possible value of x as 31.
Choice (B)

2. As the quadratic expression attains a maximum value at x = 2 the graph of the given expression is a parabola pointing downwards and a is negative.
Again, the maximum value of the expression is at $x = \frac{-b}{2a} = 2 \Rightarrow b = -4a$
∴ b is positive [∵ a is negative]
Again, the product of the two roots, $\left(\frac{c}{a}\right)$, is negative, implying c is positive.
Therefore, a is negative, whereas both b and c are positive.
Choice (C)

3. Let the initial number of workers be n and let the number of workers who were removed from the assignment at the end of every hour be t.
∴ Total number of puppets made in 4 hours and 20 minutes
$= 5n + 5(n-t) + 5(n-2t) + 5(n-3t) + 5(n-4t) \times \frac{20}{60}$
It is given that,
$5n + 5(n-t) + 5(n-2t) + 5(n-3t) + 5(n-4t) \times \frac{1}{3} = 395$
$\Rightarrow 20n - 30t + \frac{5}{3}(n-4t) = 395$
⇒ 65n – 110t = 1185
⇒ 13n – 22t = 237
∴ 13n = 237 + 22t
13n = 234 + 13t + 9t + 3
As left hand side is divisible by 13, right hand side must also be divisible by 13
Therefore 9t + 3 must be divisible by 13
The minimum value of t is 4 and the corresponding value of n is 25.
We can check that this is the only possible solution as remaining solutions for n and t give a negative value for n – 4t. [The other solutions for (n, t) are of the form (25 + 22k, 4 + 13k) for k = 1, 2, 3 …….]
Therefore, the number of puppets finished in 3 hours
= 5 × 25 + 5(25 – 4) + 5(25 – 8) = 315
Ans: (315)

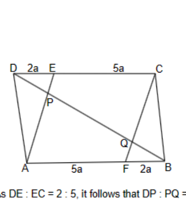
4. The number of perfect squares ≤ 10^6 is 10^3.
The number of perfect cubes ≤ 10^6 is 10^2
Of these there are common terms that are perfect sixth powers, of which there are 10 ($1^3, 4^3, 9^3, 16^3, 25^3, 36^3, 49^3, 64^3, 81^3$ and 100^3) and these appear in both the lists. Thus the number of numbers less than or equal to 10^6 which are perfect squares or cubes or fourth powers (all fourth powers are included in squares) is 1000 + 100 – 10 = 1090.
Choice (D)

5. Let the principal and the rate of interest per annum be P and r% respectively.
Total SI for 2 years = $\frac{2Pr}{100}$
Total CI for 2 years
$= P\left(1+\frac{r}{100}\right)^2 - P = \frac{2Pr}{100} + \frac{Pr^2}{100^2}$
Therefore, the difference between the total Compound Interest and the total Simple Interest for 2 years = $\frac{Pr^2}{100^2}$
= 900
Compound Interest for the 2nd year, CI_2
$= \frac{Pr}{100}\left(1+\frac{r}{100}\right) = \frac{Pr}{100} + \frac{Pr^2}{100^2} = 6900$
$\therefore \frac{Pr}{100} = 6000 \left(\because \frac{Pr^2}{100^2} = 900\right)$
$\frac{Pr}{100}\frac{r}{100} = 900$
$\Rightarrow 6000 \times \frac{r}{100} = 900$
⇒ r = 15%
$P \times \frac{15}{100} = 6000$
⇒ P = 4000
Therefore, the principal is ₹40000.

Alternative Solution:

The difference between SI and CI (for two years) is simply the interest on the interest for the first year. Since, second year's CI = 6900 and difference between SI and CI for two years = 900, we get first year's SI (or CI) = 6900 – 900 = 6000. Now, the additional 900 is the interest on 6000.
Hence, interest rate = $\frac{900}{6000}$ = 15% and principal $= \frac{6000}{0.15}$ = ₹40,000.
Ans: (40000)

6.



As DE : EC = 2 : 5, it follows that DP : PQ = 2 : 5.
As AF : FB = 5 : 2, it follows that PQ : QB = 5 : 2.
∴ DP : PQ : QB = 2 : 5 : 2 or PQ : DB = 5 : 9
As DB is 18, PQ is 10.
Ans: (10)

7. $11 - \sqrt{72} = 11 - 2\sqrt{18} = (3-\sqrt{2})^2$
Similarly, $11 + \sqrt{72} = 11 + 2\sqrt{18} = (3+\sqrt{2})^2$
$\therefore \sqrt{11-\sqrt{72}} + \sqrt{11+\sqrt{72}} = (3-\sqrt{2}) + (3+\sqrt{2}) = 6$

Alternative Solution 1:

The given expression can also be evaluated using the online calculator provided and each answer choice can then be evaluated and compared to identify the correct answer.

Alternative Solution 2:

Let $\sqrt{11-\sqrt{72}} + \sqrt{11+\sqrt{72}} = x$
Squaring on both sides, we get
$11 - \sqrt{72} + 11 + \sqrt{72} + 2\sqrt{11^2 - (\sqrt{72})^2} = x^2$
$22 + 2 \times 7 = x^2$
⇒ x = 6.
Choice (C)

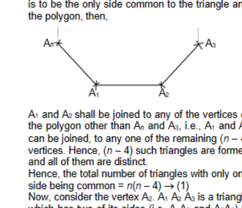
8. $(101)_{10}$ in base 8 is $(145)_8$ and $(372)_{10}$ in base 8 is $(564)_8$.
∴ $(145)_8 < (abc)_8 < (564)_8$
Again 7 > b > c > 2
Let us tabulate the values of a, b and c

a	b	c	Number of possibilities
(2/3/4/5)	4	3	4
(1/2/3/4/5)	5	3	5
(1/2/3/4/5)	6	3	5
(1/2/3/4/5)	5	4	5
(1/2/3/4)	6	4	4
(1/2/3/4)	6	5	4

Therefore, abc can assume a total of 27 possible values.
Choice (D)

9. Of the three angles one is 90° and the other two are complementary
Say, ∠A + ∠B = 90 and ∠C = 90
Now, $\sin^2A + \sin^2B + \sin^2C = \sin^2A + \sin^2(90 - B) + 1$
$= \sin^2A + \cos^2A + 1$
= 1 + 1 = 2
Ans: (2)

10. Let A_n, A_1, A_2, A_3 be any four consecutive vertices of the polygon, having n vertices. If the side A_1A_2 is to be the only side common to the triangle and the polygon, then,



A_1 and A_2 shall be joined to any of the vertices of the polygon other than A_n and A_3, i.e., A_1 and A_2 can be joined, to any one of the remaining (n – 4) vertices. Hence, (n – 4) such triangles are formed and all of them are distinct.
Hence, the total number of triangles with only one side being common = n(n – 4) → (1)
Now, consider the vertex A_2. A_1 A_2 A_3 is a triangle which has two of its sides (i.e. A_1A_2 and A_2A_3) in common with those of the polygon. Similarly, at each vertex, there is one such triangle. All these triangles are distinct.
Hence, the number of triangles, having two sides in common is n → (2)
As per the given data, n(n – 4) + n = 28
$\Rightarrow n^2 - 3n - 28 = 0$
⇒ n = 7 or –4. Ignoring the negative value, n = 7.
∴ The polygon is a heptagon.
Choice (D)

11. If we consider P = 1, 2, 3, or 4, then P! ends in zero 0's. Hence K can be 0.
Similarly for P = 5, 6, 7, 8 or 9, P! ends in one 0. Hence K can be 1.
Similarly, K can be 2, 3 or 4, but K cannot be 5, as the number of 0's at the end of 24! = 4 and the number of 0's at the end of 25! = 6. Hence, there is no value of P, for which K can be 5.
In this manner K cannot assume one value for every value of P that is a multiple of 5^2 = 25.
Similarly, K cannot assume one (additional) value for every P which is a multiple of 5^3 = 125, and one (additional) value for every P which is a multiple of 5^4 = 625.
Now, number of zeros at the end of 800! (we arrive at P = 800 by trial and error), will give 199 zero's (i.e., K < 200).
Hence, from 0 to 199, K cannot assume $\left[\frac{800}{25}\right] + \left[\frac{800}{125}\right] + \left[\frac{800}{625}\right]$ values, where [x] stands for the greatest integer less than or equal to x.
Hence, from among the 200 values possible (from 0 to 199), K can assume only 200 – (32 + 6 + 1) = 161 distinct values.
Ans: (161)

12. The amount of discount received on the first transaction
$= \frac{40}{100} \times 330 = ₹132$
The amount of discount received on the second transaction
$= \frac{40}{100} \times 370 = ₹148$
The amount of discount received on the third transaction
$= \frac{40}{100} \times 430 = ₹172 > 150$
Since, 172 > 150, the amount of discount received on third transaction is ₹150.
∴ The required percentage $= \frac{(132+148+150)}{(330+370+430)} \times 100$
$= \frac{430}{1130} \times 100 \cong 38.05\%$
Choice (D)

13. The information regarding B is not relevant here, since the ratio of A and C alone is required.
Initial ratio of A : C = 2 : 5, i.e., $\frac{1}{2.5}$. The ratio of the increases in A and C = 1 : 3, i.e., $\frac{1}{3}$.
Hence the final ratio must be less than $\frac{1}{2.5}$ (when the increase in A's earnings is negligibly small when compared to his initial earnings) but more than $\frac{1}{3}$ (when the increase in A's earnings is very large when compared to his initial earnings), i.e., if the final ratio is $\frac{1}{k}$, then 2.5 < k < 3.
Now 7 : 17 ≅ $\frac{1}{2.43}$; 9 : 22 ≅ $\frac{1}{2.44}$; 13 : 40 ≅ $\frac{1}{3.08}$
and 16 : 45 ≅ $\frac{1}{2.82}$
Hence only 16 : 45 is a possible ratio.

Alternative Solution:

Let the earnings of A and C be 2k and 5k respectively (k > 0). Let A's earnings and C's earnings increase by x and 3x respectively.
∴ Final ratio of their earnings = $\frac{2k+x}{5k+3x}$
Now, considering the ratio given in choice(A), we get
$\frac{2k+x}{5k+3x} = \frac{7}{17} \Rightarrow x = -\frac{k}{4}$, i.e., x must be negative.
This means that there was a decrease in the salaries and not an increase. This is a contradiction. Hence the final ratio of the salaries of A and C cannot be 7 : 17.
Similarly, checking for the ratios given in the other choices also, we see that only for choice(D) we get a consistent result, i.e., x is positive.
Choice (D)

14. Let the cost per kg be ₹1000.
∴ Selling price = ₹1200
But the cost is only for 800g = ₹800
Profit $= \frac{1200-800}{800} \times 100 = 50\%$
Choice (C)

15. Let the length and the breadth of the original rectangle be l and b respectively.
$\frac{l}{b} = \frac{b}{l/2}$
$2b^2 = l^2$
$\sqrt{2}b = l$
As $b = \sqrt{2}$, l = 2
Area of the smaller rectangle $= \frac{lb}{2} = \sqrt{2}$ sq.cm.
Choice (B)

16. Vinod starts at A and goes along ABCDEFA, at speed u while Raghav starts at O and goes along OABOCDO EFO with speed v. We need to consider the motion of the two, until Vinod crosses C for the second time. (i.e., if AB = r, then Vinod must cover 8r) i.e., until he completes one revolution and reaches C in his second revolution.
Raghav has to overtake Vinod on AB, CD and EF.
If he overtakes Vinod on AB in the first round v ≥ 2u. (If that is so, when Vinod is at C (having covered 2r) and Raghav is at C (having covered 4r or beyond). If he has to overtake Vinod on CD, v = 2u.
But when Vinod is at E (having covered 4r) Raghav is at F (having covered 8r) and it is not possible for Raghav to overtake Vinod on EF.
∴ Raghav doesn't overtake Vinod on AB in the first round. He does so in the second round.
The distance that Vinod covers and the bounds (minimum and maximum) of the distances that Raghav covers are tabulated below.

Distance covered by Vinod	Min. distance covered by Raghav	Max. distance covered by Raghav
r	2r	
2r		4r
3r	5r	
4r		7r
5r	8r	
6r		10r
7r	11r	

From row 3, v ≥ 5/3 u
From row 6, v ≤ 5/3 u
∴ v = 5/3 u
Choice (D)

Explanation:

Row 3: In order that Raghav overtakes Vinod on CD, he has to cover at least 5r (OA + AB + BP + OC + CD) by the time Vinod covers 3r (AB + BC + CD)
Similarly, the rest of the table can be completed.

17. Let the number of brown sarees be x. let the number of shirts not having a design be y.

Shirts		Sarees			
Having Design	3y	Yellow / Black	5x / 2x	Hand spun	4k
Not having Design	y	Brown	x	Not Hand spun	k
Total	4y		8x		5k

Hence total number of sarees, say N = 8x = 5k
The least such number is 8 × 5 = 40.
Given 4y + N = 72
⇒ N = 40 itself and 4y = 32
⇒ Shirts not having design = y = 8.
Choice (B)

18. Let the price per kg of the three varieties be p_1, p_2 and p_3.
Let the quantities of the three varieties purchased be q_1, q_2 and q_3.
$\therefore \frac{q_1p_1 + q_2p_2}{q_1+q_2} = 50 \Rightarrow q_1p_1 + q_2p_2 = 50(q_1 + q_2)$ –(1)
Similarly $q_2p_2 + q_3p_3 = 60(q_2 + q_3)$ ----- (2) and
$q_3p_3 + q_1p_1 = 68(q_3 + q_1)$ ----- (3)
Adding (1), (2) and (3), we get
$\Rightarrow 2(q_1p_1 + q_2p_2 + q_3p_3) = 118q_1 + 110q_2 + 128q_3$
$\Rightarrow q_1p_1 + q_2p_2 + q_3p_3 = 59q_1 + 55q_2 + 64q_3$
Hence, $\frac{(q_1p_1 + q_2p_2 + q_3p_3)}{(q_1+q_2+q_3)}$ lies between 55 and 64 (both exclusive).
Choice (D)

19. Since for the first 20% of the distance, he drove at normal speed, the time taken = 20%(150) = $\frac{1}{5}$(15)
= 30 minutes. The time remaining, if he had continued at his normal speed = 150 – 30 = 120 min.
But it is given that the remaining distance was covered in 190 – 30 = 160 minutes.
That is, the time remaining became 4/3 times the original remaining time. This implies that the speed (after a decrease of 15 kmph) became 3/4th of the original. That is 15 kmph = $\left(\frac{1}{4}\right)^{th}$ of original speed.
Hence, original speed = $\frac{15}{\left(\frac{1}{4}\right)}$ = 60 kmph
Therefore, the total distance between his house and his in–law house
$= 60 \text{ kmph} \times \left(\frac{150}{60}\right)$ hrs
= 150 km
Ans: (150)

20. The amount to be paid by Pradeep to bank A after n years $= 10{,}00{,}000\left(1+\frac{20}{100}\right)^n$
The amount to be paid by Pradeep to bank B after n years $= 10{,}00{,}000\left(1+\frac{25n}{100}\right)$
∴ We have to maximise 'n' so that
$\left(1+\frac{20}{100}\right)^n < 1 + \frac{25n}{100}$
$\Rightarrow (1.20)^n < 1 + 0.25n$
For n = 4, $(1.20)^n > 1 + 0.25n$.
Hence, maximum value of n for bank A to be a better option is 3.
Choice (B)

21. It is given that, $T_5 = 11$ and $T_{16} = 19$
$\therefore S_{20} = \frac{[T_1 + T_{20}]}{2} \times 20 = \frac{[T_5 + T_{16}]}{2} \times 20$
= 10 [11 + 39] = 500

Alternative Solution:

Let a, d be the first term and the common difference respectively.
a + 4d = 11
a + 15d = 39
Solving for a and d, we get $a = \frac{9}{11}$ and $d = \frac{28}{11}$
Sum of the first 20 terms $= \frac{20}{2}(2a + 19d) = 500$.
Ans: (500)

22. 12, 345, 678, 987, 654, 321 = N
$= 12 \times 10^{15} + 345 \times 10^{12} + 678 \times 10^9 + 987 \times 10^6 + 654 \times 10^3 + 321$
$\frac{1000}{1001}$ leaves a remainder of 1000 or –1.
∴ The given number is expressed in the above form.
$\text{Rem}\left(\frac{10^3}{1001}\right) = -1$ $\text{Rem}\left(\frac{10^6}{1001}\right) = 1$
$\therefore \text{Rem}\left(\frac{12 \times 10^{15} + 345 \times 10^{12} + 678 \times 10^9 + 987 \times 10^6 + 654 \times 10^3 + 321}{1001}\right)$
= 12(–1) + 345(1) + 678(–1) + 987(1) + 654(–1) + 321
= –12 + 345 – 678 + 987 – 654 + 321 = 333 + 309 – 333
= 309.

Alternative Solution:

Though time consuming, it is possible to directly approach the question using the long division method to arrive at the remainder when the given number is divided by 1001.
Ans: (309)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	7
Easy	9, 12, 14, 21
Medium	1, 2, 4, 5, 6, 8, 10, 13, 15, 17, 18, 19, 20
Difficult	3, 16, 22
Very Difficult	11

Done
Tip: Scroll slowly for the best results